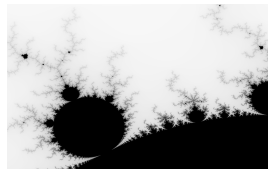


Mandelbrot set

And where is Mandelbrot in all this?

- How are we going to compute the Mandelbrot set?
- We're just going to break the formula down.



$$S_m = \begin{cases} z_0 = 0 & \text{initial state} \\ z_{n+1} = z_n^2 + c & \text{we iterate to infinity} \end{cases}$$

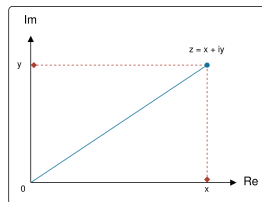
- In our particular case, we're going to compute up to **100 iterations**.

Computation of the Mandelbrot set

One iteration $z_{n+1} = z_n^2 + c$

With $z = z_{re} + \imath z_{im}$, and $c = c_{re} + \imath c_{im}$

Divergence condition $|z| < 2, z \in S_m$



Computation of the Mandelbrot set



$$z_{n+1} = z_n^2 + c$$

Computation of the Mandelbrot set

-

$$z_{n+1} = z_n^2 + c$$

-

$$z_{n+1} = (z_{re_n} + \imath z_{im_n})^2 + c_{re_n} + \imath c_{im_n}$$

Computation of the Mandelbrot set

- $$z_{n+1} = z_n^2 + c$$

- $$z_{n+1} = (z_{re_n} + \imath z_{im_n})^2 + c_{re_n} + \imath c_{im_n}$$

- $$z_{n+1} = z_{re_n}^2 + 2 * z_{re_n} * \imath z_{im_n} + (\imath z_{im_n})^2 + c_{re_n} + \imath c_{im_n}$$

Computation of the Mandelbrot set

- $$z_{n+1} = z_n^2 + c$$

- $$z_{n+1} = (z_{re_n} + \imath z_{im_n})^2 + c_{re_n} + \imath c_{im_n}$$

- $$z_{n+1} = z_{re_n}^2 + 2 * z_{re_n} * \imath z_{im_n} + (\imath z_{im_n})^2 + c_{re_n} + \imath c_{im_n}$$

- $$z_{n+1} = (z_{re_n}^2 - z_{im_n}^2 + c_{re_n}) + \imath (2 * z_{re_n} * z_{im_n} + c_{im_n})$$

Computation of the Mandelbrot set

- $$z_{n+1} = z_n^2 + c$$

- $$z_{n+1} = (z_{re_n} + \imath z_{im_n})^2 + c_{re_n} + \imath c_{im_n}$$

- $$z_{n+1} = z_{re_n}^2 + 2 * z_{re_n} * \imath z_{im_n} + (\imath z_{im_n})^2 + c_{re_n} + \imath c_{im_n}$$

- $$z_{n+1} = (z_{re_n}^2 - z_{im_n}^2 + c_{re_n}) + \imath (2 * z_{re_n} * z_{im_n} + c_{im_n})$$

- $$\begin{cases} z_{re_{n+1}} = z_{re_n}^2 - z_{im_n}^2 + c_{re_n} \\ z_{im_{n+1}} = 2 * z_{re_n} * z_{im_n} + c_{im_n} \end{cases}$$

Computation of the Mandelbrot set

$$z_{n+1} = z_n^2 + c$$

$$z_{n+1} = (z_{re_n} + \imath z_{im_n})^2 + c_{re_n} + \imath c_{im_n}$$

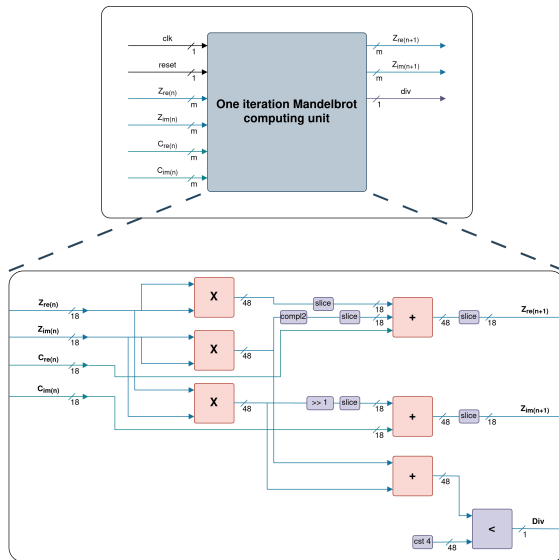
$$z_{n+1} = z_{re_n}^2 + 2 * z_{re_n} * \imath z_{im_n} + (\imath z_{im_n})^2 + c_{re_n} + \imath c_{im_n}$$

$$z_{n+1} = (z_{re_n}^2 - z_{im_n}^2 + c_{re_n}) + \imath (2 * z_{re_n} * z_{im_n} + c_{im_n})$$

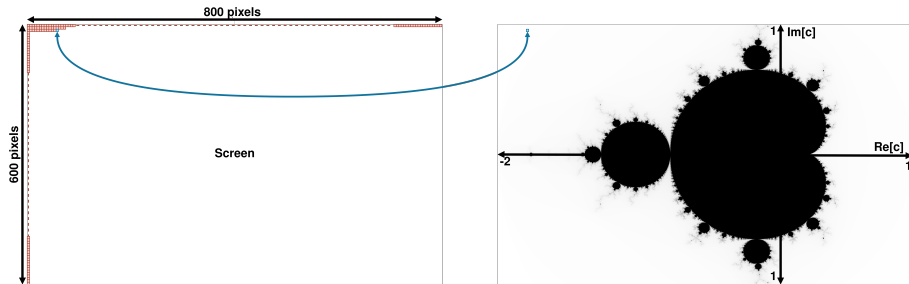
$$\begin{cases} z_{re_{n+1}} = z_{re_n}^2 - z_{im_n}^2 + c_{re_n} \\ z_{im_{n+1}} = 2 * z_{re_n} * z_{im_n} + c_{im_n} \end{cases}$$

$$|z_n| < 2 \implies \sqrt{z_{re_n}^2 + z_{im_n}^2} < 2 \text{ or } z_{re_n}^2 + z_{im_n}^2 < 4$$

Computation of the Mandelbrot set



Computation of the Mandelbrot set



$$step_{horiz} = \frac{3}{800 - 1} = 0.003754693 \text{ and } step_{vert} = \frac{2}{600 - 1} = 0.003338898$$

$$c_{(13,3)} = \begin{cases} c_{re(13,3)} = 0.003754693 * 13 = 0.048811009 \\ c_{im(13,3)} = 0.003338898 * 3 = 0.010016694 \end{cases}$$

$$c_{(13,3)} = 0.048811009 + i * 0.010016694 \text{ Oops, we have real numbers...}$$

To make your life easier, we're going to use fixed point numbers

- Fixed point numbers **are used as integers**.
- But you must **anticipate the size of the integer part**.
- For multiplication, **the dynamics of the numbers** must be **doubled**.
- Be careful, **the commas** must be **aligned**.
- **Let's get started.**

